Statistics

Chapter 12 — Instructor Solutions Manual

October 24, 2026 • Total Exercises: 3

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 18$$
$$\overline{x} = \text{Sample mean} = 96$$
$$s = \text{Sample standard deviation} = 12$$

Given claim: Mean is greater than 90

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 90$$
$$H_a : \mu > 90$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all t-value above the t-value t_0 that has a probability of 0.01 to its right.

$$P(t > t_0) = 0.01$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 12 - 1 = 11$ and in the column with $\alpha = 0.01$.

$$t = 2.718$$

The rejection region then contains all values greater than 2.718 and thus we reject H_0 when the t-value is larger than 2.718.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{96 - 90}{12/\sqrt{18}} \approx 2.121$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = 2.121 < 2.718 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the mean time in the park is greater than 90 minutes.

Result

Not reasonable

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

Given:

$$n_1 = \text{Sample size} = 14$$
$$n_2 = \text{Sample size} = 12$$
$$\overline{x}_1 = \text{Sample mean} = 837$$
$$\overline{x}_2 = \text{Sample mean} = 797$$
$$s_1 = \text{Sample standard deviation} = 30$$
$$s_2 = \text{Sample standard deviation} = 40$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value t_0 that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 14 + 12 - 2 = 24$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.064$$

The rejection region then contains all values smaller than -2.064 and all values greater than 2.064,thus we reject H_0 when $t < -2.064$ or $t > 2.064$.

Step 3

Determine the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(14 - 1)30^2 + (12 - 1)40^2}{14 + 12 - 2}} \approx 34.9404$$

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{837 - 797}{34.9404\sqrt{\frac{1}{14} + \frac{1}{12}}} \approx 2.910$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 2.910 > 2.064 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean miles traveled.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 12, Page 460

Solution A

Step 1

(a) The degrees of freedom for the treatments is given in the row "Between groups" and in the column "df" of the given ANOVA table:

$$d.f._T = 2$$

The degrees of freedom for the treatments is the number of treatments decreased by 1, which implies that the number of treatments is the degrees of freedom for the treatments increased by 1.

$$\text{Number of treatments} = d.f._T + 1 = 2 + 1 = 3$$

Step 2

(b) The total degrees of freedom is given in the row "Total" and in the column "df" of the given ANOVA table:

$$d.f._{total} = 20$$

The total degrees of freedom is the total sample size decreased by 1, which implies that the total sample size is the total degrees of freedom increased by 1.

$$n = \text{Total sample size} = d.f._{total} + 1 = 20 + 1 = 21$$

Step 3

(c) The degrees of freedom for the treatments is given in the row "Between groups" and in the column "df" of the given ANOVA table:

$$d.f._T = 2$$

The degrees of freedom for the error is given in the row "Within groups" and in the column "df" of the given ANOVA table

$$d.f._E = 18$$

The F- critical value is given in the row with $dfd = d.f._E = 18$ and in the column with $dfn = d.f._T = 2$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 3.55$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 3.55.

Step 4

(d) The null hypothesis states that all population means are equal:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The alternative hypothesis states the opposite of the null hypothesis:

$$H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$$

Step 5

(e) The value of the test statistic is given under "F" in the ANOVA table:

$$F = 4.960047$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$F = 4.960047 > 3.55 \Rightarrow \text{ Reject } H_0$$

Step 6

(e) Since we rejected the null hypothesis in part (e):

There is sufficient evidence to support the claim that the treatment means differ.

Result

(a) 3 treatments

(b) $n = 21$

(c) $F = 3.55$

(d) $H_0 : \mu_1 = \mu_2 = \mu_3, H_1 : \text{ not all of } \mu_1, \mu_2, \mu_3 \text{ are equal}$

(e) Reject H_0

(f) Yes